

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	27.0 / Female
Department	Gynaecology
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Burning urination;Pelvic pain

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	39.0	%	20 - 40
Wbc	8850.0	cells/μL	4000 - 11000
Polymorphs	62.0	%	40 - 75
Crp	8.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.1	mg/dL	3.5 - 7.2
Blood Urea	24.0	mg/dL	7 - 20
Cue Pus Cells	11.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Enterococcus faecium
Bacteria Type	Gram-positive coccus (Higher risk of Vancomycin resistance - VRE)

Antibiotic Susceptibility Profile

Predicted Sensitive	Daptomycin
Predicted Resistant	Clindamycin

Recommended Therapy

Daptomycin

Dosage: 6 mg/kg IV once daily (adjusted for body weight)

Precautions: Normal renal function; monitor serum creatine kinase for eosinophilic pneumonia; avoid in patients with known hypersensitivity; no liver impairment noted; not typically used for urinary tract infections due to low urinary excretion, but used here per sensitivity data.

Rationale: Daptomycin is active against Enterococcus faecium, including vancomycin-resistant strains; the recommended dose is 6 mg/kg IV daily for 10-14 days in uncomplicated infections.

Antibiotic Drug History

Clinical Summary

Clinical Summary

- **Infection & Organism:** Recurrent urethritis caused by *Enterococcus faecium*, a Gram-positive coccus with a high risk of vancomycin resistance (VRE).
- **Resistance/Sensitivity Profile:** The isolate is predicted resistant to clindamycin (and likely vancomycin) and sensitive to daptomycin.
- **Recommended Therapy:**
 - **Daptomycin 6 mg/kg IV once daily** (adjusted for body weight).
 - Duration: 10-14 days for uncomplicated urethritis.
 - *Rationale:* Daptomycin retains activity against VRE strains, providing a reliable option when clindamycin and vancomycin are ineffective.
- **Key Precautions & Considerations:**

- Normal renal function (serum creatinine 0.7 mg/dL) - no dose adjustment needed.
- Monitor serum creatine kinase (CK) daily; elevated CK may indicate eosinophilic pneumonia.
- Avoid in patients with known hypersensitivity to daptomycin.
- Daptomycin has low urinary excretion, so it is not first-line for UTIs; it is chosen here solely because of in vitro sensitivity and lack of alternative options.
- No hepatic impairment noted; routine liver function monitoring is not required.
- **Clinical Context:** The patient is a 27-year-old woman with no comorbidities or renal dysfunction. Prior antibiotics (clindamycin, vancomycin) have been used with limited success; clindamycin resistance is confirmed, and vancomycin likely ineffective due to VRE.

This regimen offers a targeted, evidence-based approach to eradicate the VRE urethritis while minimizing toxicity risks.